Bike Stations Sharing Analysis

Project Objective

To analyze bike-sharing station availability, identify stations with high and low bike availability, understand contract bike distribution, and help improve bike distribution and availability

Data Sources

Source: Dataset link

Timeline: January 2022 to November 2025

Domain: Sales Rental

Problem Statement

- To analyse the highest and lowest number of available bikes in stations
- To analyse the bonus and banking influence in bike stands
- To analyse How many bike stations are open and closed
- To understand bike stands distribution by contractors

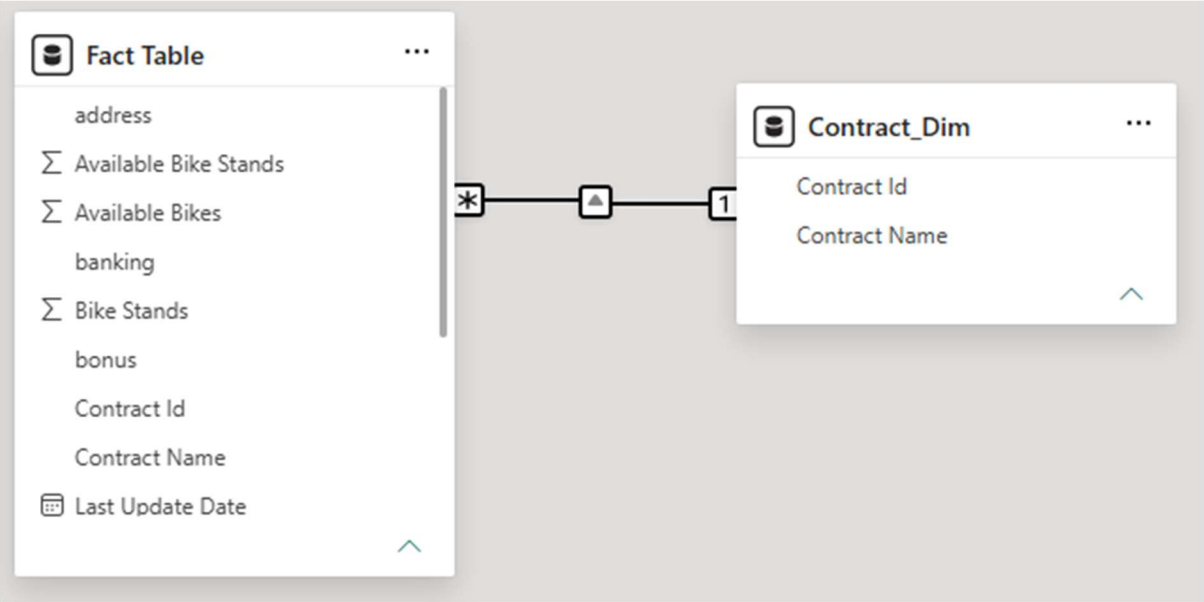
Attribute Details

Attribute Name	Description
Number	Unique identification of bike station
Name	Name of bike station
Address	Address of bike station
Position 1	Latitude position of bike station address
Position 2	Longitude position of bike station address
Banking	Is banking option available or not
Bonus	Shows bonus available or not
Status	Shows open or closed status of the station
Contract Name	Particular contract name
Contract Id	Unique contract id of contractors
Bike Stands	Total bike stands of the city
Available Bike Stands	Currently available bike stands
Last Update Date	Shows last updated date
Last Update Time	Shows last updated date

Tools & Technologies

- Excel: Create dimension table from fact table
- Power BI: Data cleaning, Transformation, Data Modelling, DAX calculations, Visualizations, Interactive Dashboard creation

Data Modelling



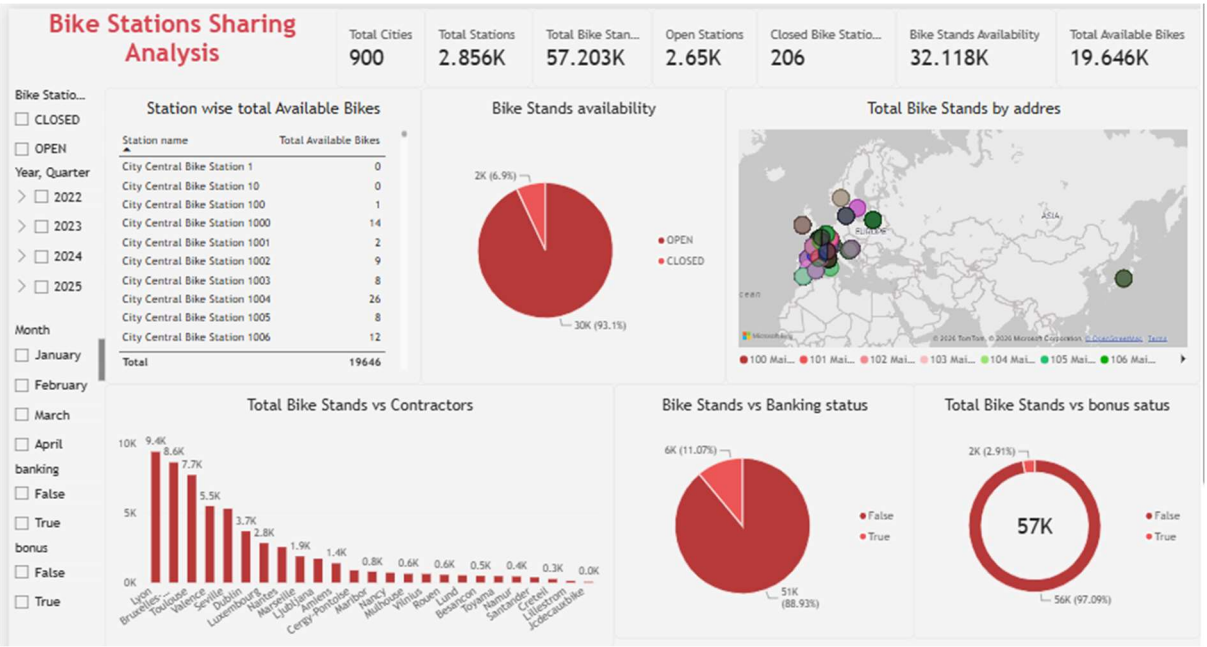
Dax Measures

- Total Stations=COUNT(Fact Table[number])
- Total Cities=DISTICNTCOUNT(Fact Table[address])
- Total Bike Stands=sum(Fact Table[Bike Stands])
- Bike Stands Availability=sum(Fact Table[Available Bike Stands])
- Total Available Bikes=sum(Fact Table[Available Bikes])
- Open Stations=CALCULATE(Total Stations, 'Fact Table'[status]="OPEN"])
- Closed Stations=CALCULATE(Total Stations, 'Fact Table'[status]="OPEN"])

Analysis & Visualizations

- Used Table chart for understand city wise total available bikes distribution
- Used pie chart to analyse percentage of open and closed bike stands
- And also calculate the percentage of bike stands vs banking status availability
- Used Geographical map to show no of bike stands distributed in cities using positions
- Used column chart to understand contract wise total bike stands
- Used donut chart for calculate the percentage of bonus status of bike stands

Report



Insights

- Totally 900 different cities are having the bike stations
- Totally 2.85k stations are available
- Open stations 2.65k
- Closed stations 206
- Total bike stands 57.203k
- Total bike stand availability is 32.118k
- Totally 19.646k bikes are available
- Maximum of 93.1% bike stands are open
- Only 11.07% of bike stands are having banking support
- Only 2.91% of bike stands with true bonus status
- Lyon contract have highest number of bikes stands totally 9350

Analysis Insights

- The station-wise available bikes analysis helps identify how bike availability varies from one station to another.
- The total number of cities and bike stations provides an overview of the geographical coverage of the bike-sharing network
- Station status analysis highlights the distribution of open and closed bike stations.
- The comparison of bike stands with banking status helps understand how banking facilities are distributed across stations.
- The comparison of bike stands with bonus status helps identify how bonus-enabled facilities are distributed across the bike-sharing network

- Geographic distribution and station capacity can help identify locations where future infrastructure expansion may be required
- Prioritize stations that are closed or frequently experience operational issues.
- Review banking and bonus facilities to identify opportunities to improve station services.

Conclusion

The Bike Stations Sharing Analysis project provides a comprehensive overview of the bike-sharing network, focusing on station distribution, bike availability, operational status, banking & bonus status and geographical coverage